

Theorem 1. For every x ,

$$(((x \setminus e) \setminus e) \setminus x) \setminus x = ((((((x \setminus e) \setminus e) \setminus x) \setminus x) \setminus e) \setminus e).$$

Proof. Introduce the abbreviations

$$u = x \setminus e, \quad v = u \setminus e, \quad w = v \setminus x,$$

and

$$a = w \setminus x, \quad b = a \setminus e, \quad c = b \setminus e.$$

It is therefore enough to prove $a = c$.

First, the contraction and mingle axioms give

$$p \setminus (p \setminus q) \leq p \setminus q \quad \text{and} \quad p \setminus q \leq p \setminus (p \setminus q).$$

Hence, by antisymmetry,

$$p \setminus (p \setminus q) = p \setminus q. \tag{1}$$

We shall also use the following consequence of the axioms:

$$p \leq q \implies q \setminus e \leq p \setminus e. \tag{2}$$

Indeed, from

$$q \leq (q \setminus e) \setminus e$$

and $p \leq q$, transitivity gives

$$p \leq (q \setminus e) \setminus e.$$

Using $r \leq s \iff e \leq r \setminus s$, followed by exchange, we obtain

$$e \leq p \setminus ((q \setminus e) \setminus e) = (q \setminus e) \setminus (p \setminus e).$$

Applying the same order characterization once more proves (2).

The axiom $p \leq (p \setminus q) \setminus q$, with $p = a$ and $q = e$, gives

$$a \leq (a \setminus e) \setminus e = c. \tag{3}$$

We next prove

$$a = w \setminus v. \tag{4}$$

The axiom

$$p \setminus q \leq (q \setminus r) \setminus (p \setminus r)$$

with $p = w, q = x, r = e$ gives

$$a = w \setminus x \leq (x \setminus e) \setminus (w \setminus e) = u \setminus (w \setminus e).$$

By exchange,

$$u \setminus (w \setminus e) = w \setminus (u \setminus e) = w \setminus v,$$

so $a \leq w \setminus v$.

Conversely, applying the same axiom with $p = w, q = v, r = x$ gives

$$w \setminus v \leq (v \setminus x) \setminus (w \setminus x) = w \setminus a.$$

Since $a = w \setminus x$, equation (1) yields

$$w \setminus a = w \setminus (w \setminus x) = w \setminus x = a.$$

Thus $w \setminus v \leq a$, proving (4) by antisymmetry.

From (3) and (2),

$$c \setminus e \leq a \setminus e = b.$$

On the other hand,

$$b \leq (b \setminus e) \setminus e = c \setminus e.$$

Consequently,

$$b = c \setminus e. \tag{5}$$

Using exchange, first with $b = a \setminus e$ and then with $b = c \setminus e$, we obtain

$$u \setminus b = u \setminus (a \setminus e) = a \setminus (u \setminus e) = a \setminus v$$

and

$$u \setminus b = u \setminus (c \setminus e) = c \setminus (u \setminus e) = c \setminus v.$$

Therefore

$$a \setminus v = c \setminus v. \tag{6}$$

Since $a = w \setminus v$, exchange gives, for every t ,

$$t \setminus a = t \setminus (w \setminus v) = w \setminus (t \setminus v).$$

Taking $t = a$ and $t = c$, and using (6), gives

$$a \setminus a = w \setminus (a \setminus v) = w \setminus (c \setminus v) = c \setminus a.$$

The axiom $e \leq a \setminus a$ therefore implies

$$e \leq c \setminus a.$$

By $p \leq q \iff e \leq p \setminus q$, this means

$$c \leq a.$$

Together with (3), antisymmetry yields $a = c$, which is precisely the required identity. $\square$